TITANIC SURVIVAL PREDICTION

Random Forest Classifier | Project Report

1. Dataset Name

Kaggle – *Titanic: Machine Learning from Disaster* (**train.csv**). The dataset contains 891 passenger records and 12 columns. The target column is **Survived**: 0 means the passenger did not survive and 1 means the passenger survived. There are 342 survivors (38.4%) and 549 non-survivors (61.6%). Source: https://www.kaggle.com/c/titanic

2. Problem Statement

The aim of this project is to build a machine learning model that predicts whether a Titanic passenger survived. The model uses details such as age, sex, passenger class, fare and family information to make a binary prediction.

3. Data Preprocessing Steps

Task	Preprocessing performed
Missing values	Age was filled using the median for each Pclass–Sex group. The two missing Embarked values were filled with the mode. As around 77% of Cabin values were missing, Cabin was changed into a HasCabin indicator and the original column was removed.
Outliers	Fare values were capped at the 99th percentile to reduce the influence of unusually high fares.
Feature engineering	Title was extracted from Name. FamilySize was calculated as SibSp + Parch + 1, and IsAlone was created to identify passengers travelling alone.
Encoding and selection	Sex, Embarked and Title were label-encoded. PassengerId, Name and Ticket were dropped from the model features.
Data split	The data was divided into 80% training and 20% testing sets using stratify=y to keep the survival proportion similar in both sets.

4. Model Used

A **Random Forest Classifier** with 200 decision trees was used. First, a baseline model was trained. Then, GridSearchCV with 5-fold cross-validation and F1 scoring was used to find better hyperparameters. The selected settings were max_depth=10, min_samples_leaf=1, min_samples_split=10 and n_estimators=200. The final model was saved in a .joblib file.

5. Evaluation Metrics

Metric	Result
Baseline accuracy	Training: 98.74% | Testing: 79.89%
Tuned model	Accuracy: 78.21% | Precision: 0.7206 | Recall: 0.7101 | F1-score: 0.7153
5-fold cross-validation	Mean accuracy: 82.94% (± 0.0267)
Confusion matrix	TN: 91 | FP: 19 | FN: 20 | TP: 49

6. Key Findings

The baseline model showed overfitting because its training accuracy (98.74%) was much higher than its testing accuracy (79.89%). After tuning, the reported training accuracy reduced to 90.87%, indicating a smaller gap. The tuned model achieved 78.21% test accuracy. Feature importance identified **Sex (25.3%)**, **Fare (17.7%)** and **Title (14.5%)** as the top three predictors. This suggests that gender and socio-economic factors played an important role in the model's predictions.

7. Conclusion

Through this project, I learned how to prepare a real-world dataset, create useful features, train a Random Forest model and evaluate its performance. Hyperparameter tuning helped reduce the overfitting seen in the baseline model. The 5-fold cross-validation accuracy was 82.94%, and the final model was saved using Joblib. In the future, I can compare this model with Gradient Boosting methods to see whether the results can be improved.

Note: Dataset dimensions and target counts were checked from the uploaded train.csv. Model metrics and feature-importance values are reproduced from the results provided; they have not been independently recalculated.